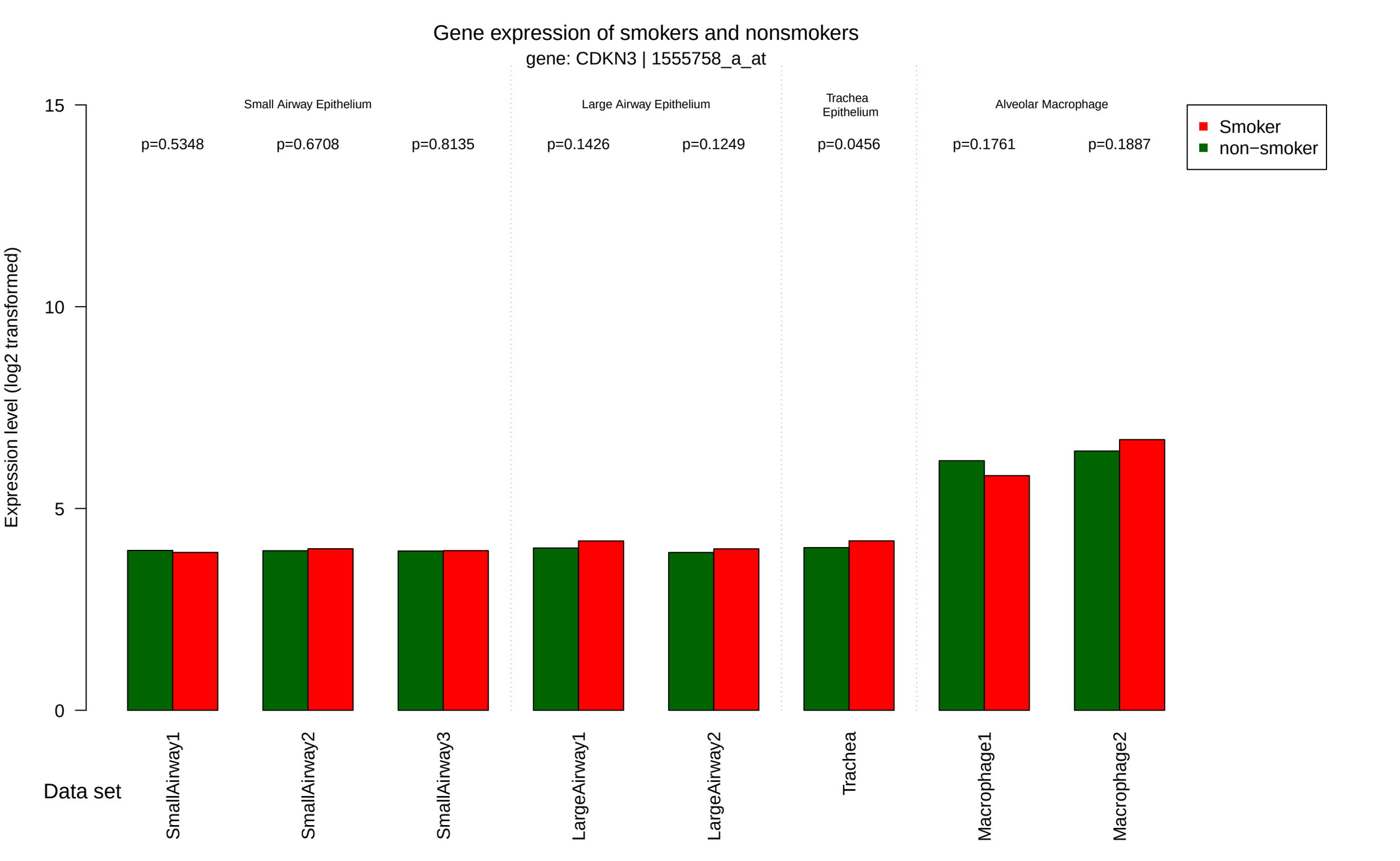
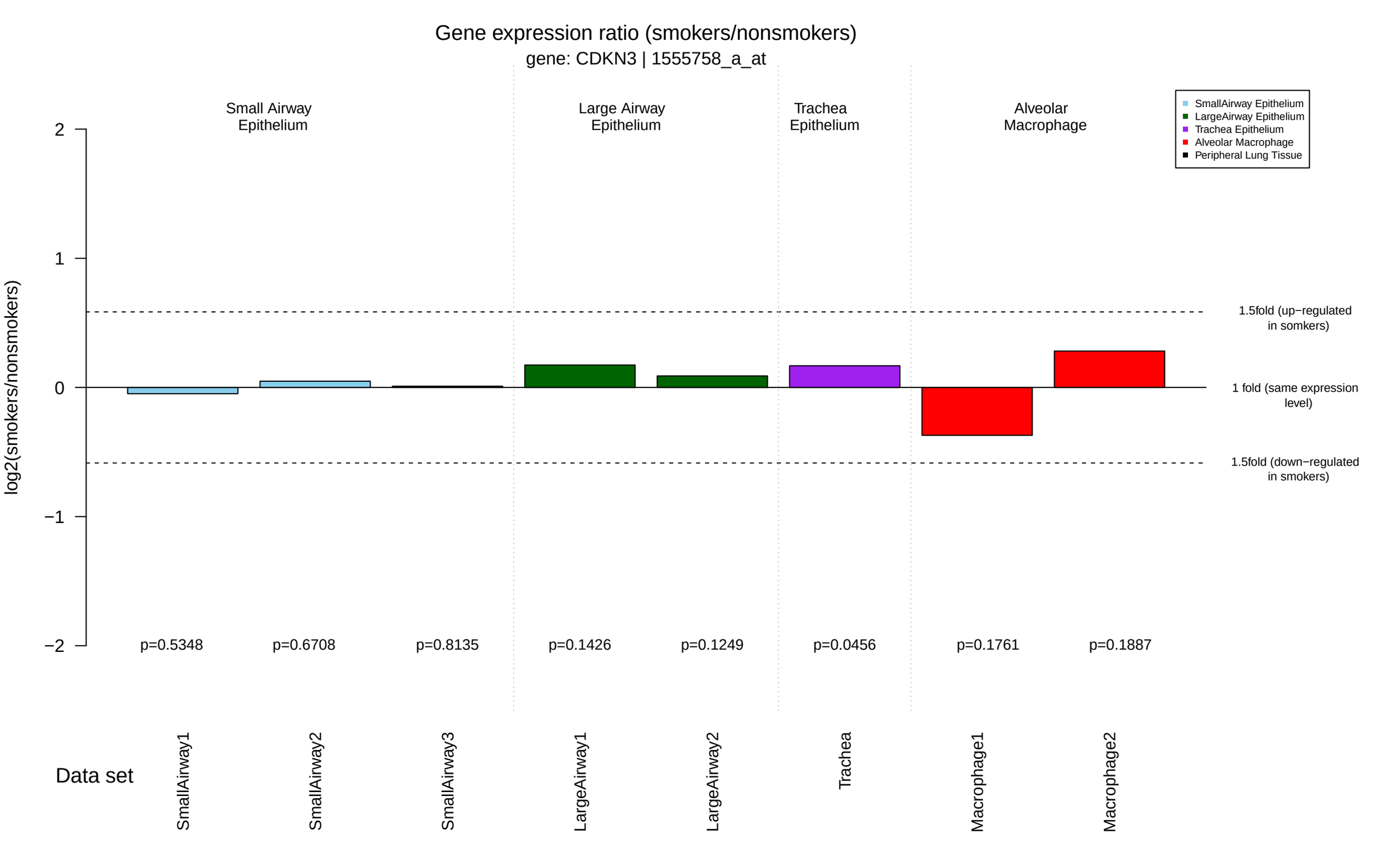
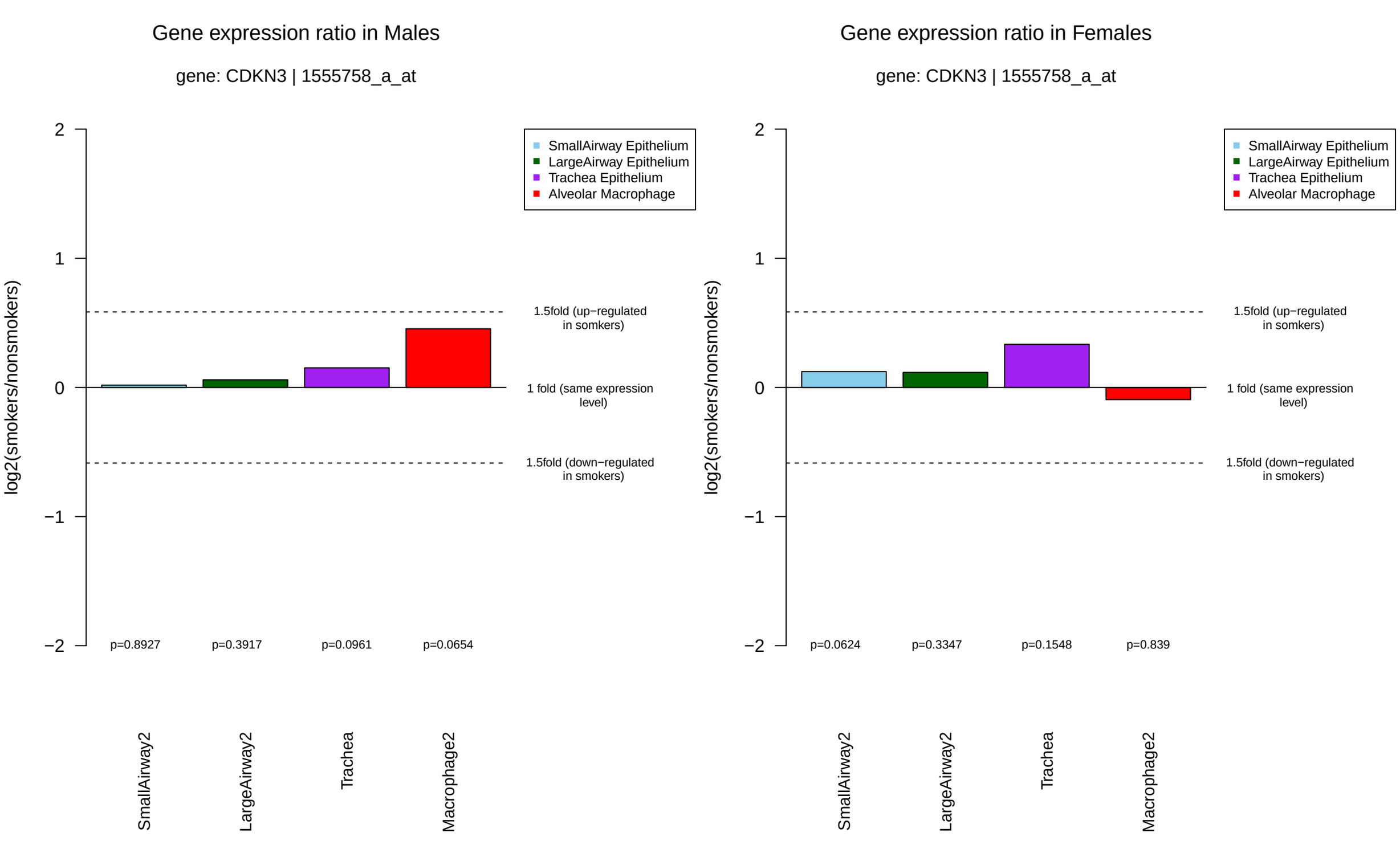
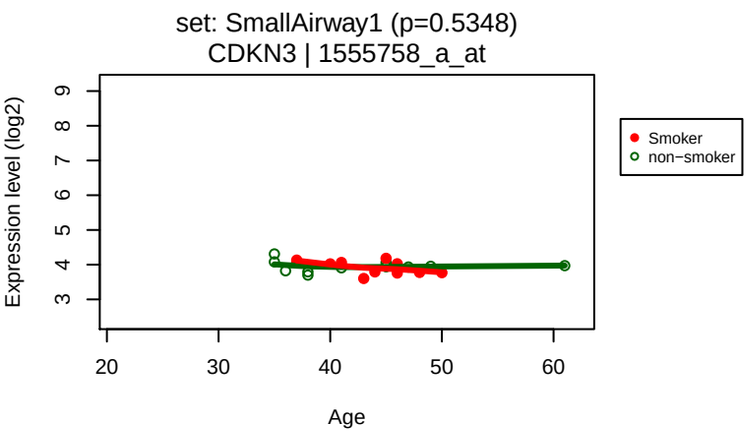
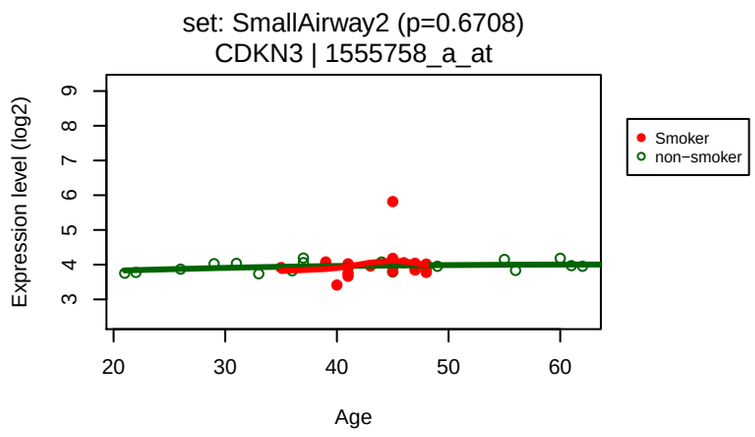
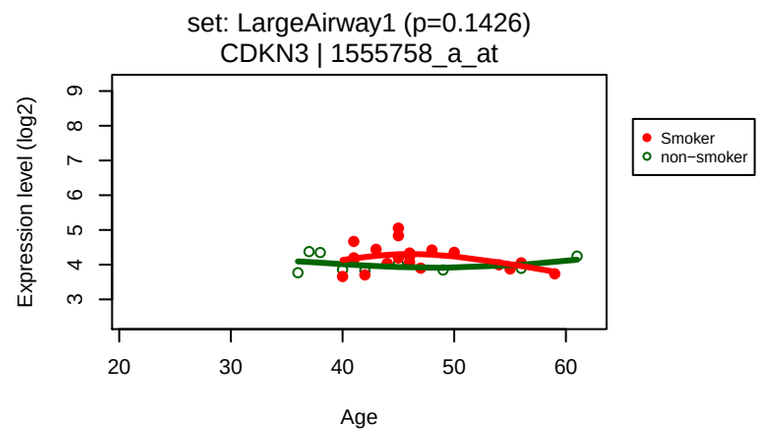

Gene expression of smokers and nonsmokers
gene: CDKN3 | 1555758_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5348
p=0.6708
p=0.8135
p=0.1426
p=0.1249
p=0.0456
p=0.1761
p=0.1887
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

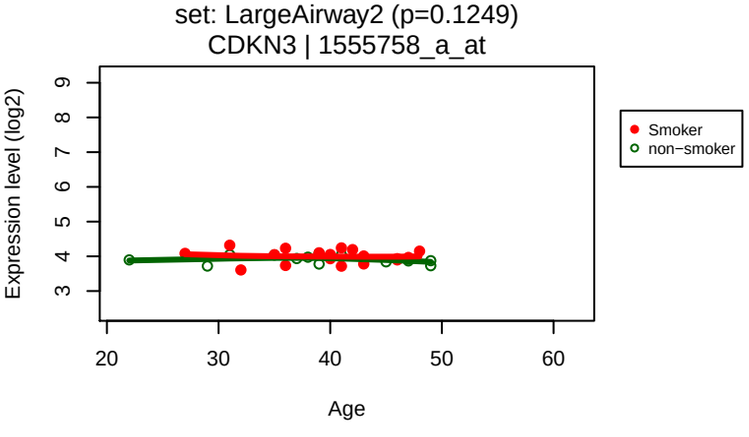

Gene expression ratio (smokers/nonsmokers)
gene: CDKN3 | 1555758_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5348
p=0.6708
p=0.8135
p=0.1426
p=0.1249
p=0.0456
p=0.1761
p=0.1887
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

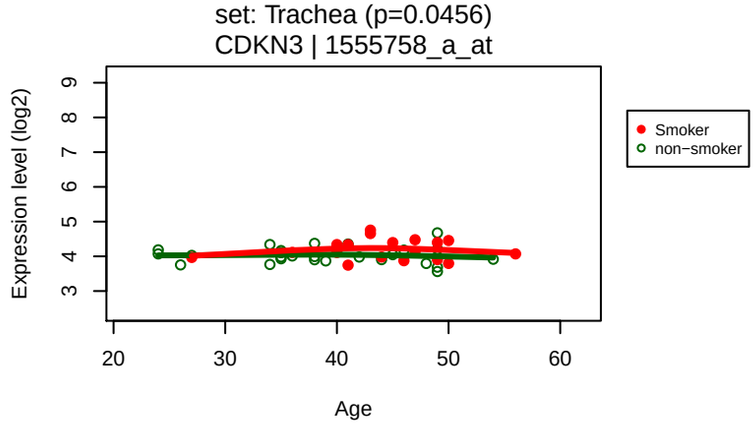

Gene expression ratio in Males
gene: CDKN3 | 1555758_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8927
p=0.3917
p=0.0961
p=0.0654
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDKN3 | 1555758_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0624
p=0.3347
p=0.1548
p=0.839
SmallAirway2
LargeAirway2
Trachea
Macrophage2

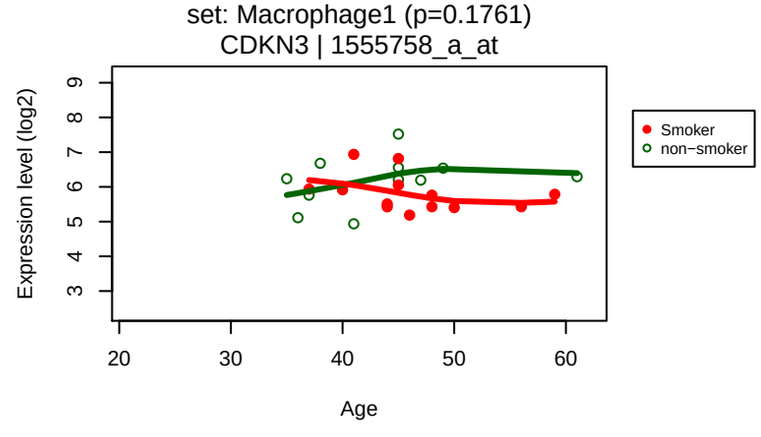

set: SmallAirway1 (p=0.5348)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker

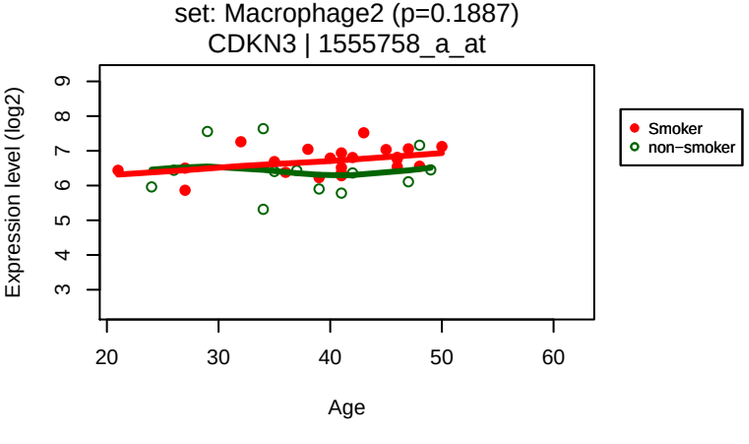

set: SmallAirway2 (p=0.6708)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker

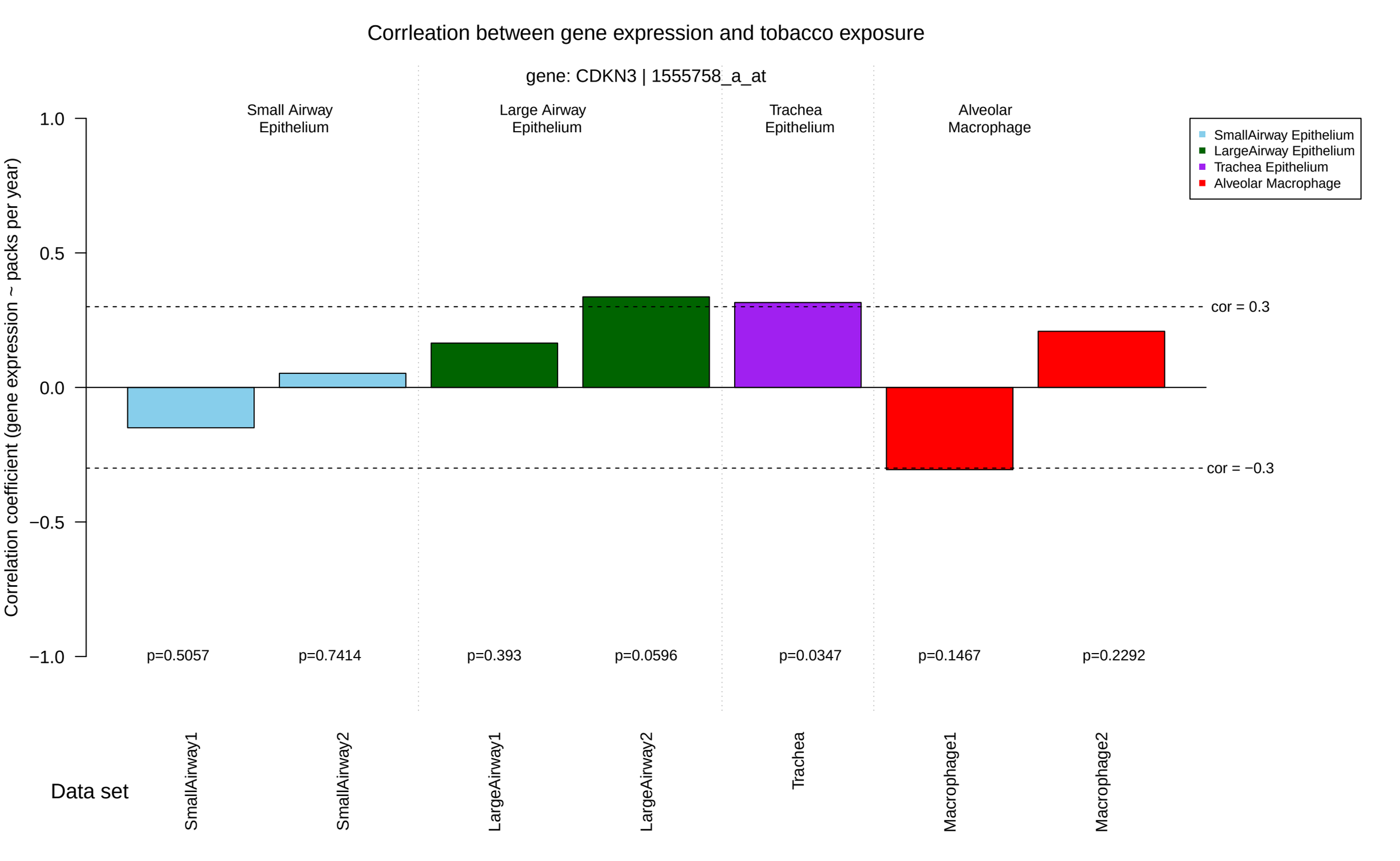

set: LargeAirway1 (p=0.1426)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1249)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0456)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1761)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1887)
CDKN3 | 1555758_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDKN3 | 1555758_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5057
p=0.7414
p=0.393
p=0.0596
p=0.0347
p=0.1467
p=0.2292
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2